

Comparative Analysis of Machine Learning Models for RDF-Based Mutagenicity Prediction

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Abstract

This project presents a machine learning framework for predicting the mutagenicity of chemical compounds using RDF-derived molecular features. The extracted features are integrated with the MUTAG dataset to train and evaluate Logistic Regression, Support Vector Machine (SVM), and Decision Tree classifiers. Model performance is assessed using Accuracy, Precision, Recall, and F1-score, while interpretability is analysed through coefficient analysis, feature weights, and feature importance. The results demonstrate that RDF-derived semantic features provide an effective and interpretable representation for mutagenicity prediction.

1 Introduction

Mutagenicity prediction is an important task in cheminformatics and computational toxicology, supporting drug discovery and chemical safety assessment. Machine learning provides an efficient alternative to experimental testing by predicting whether chemical compounds are mutagenic based on their molecular characteristics.

In this project, molecular features extracted from a Resource Description Framework (RDF) knowledge graph are combined with the MUTAG benchmark dataset to build binary classification models. Logistic Regression, Support Vector Machine (SVM), and Decision Tree classifiers are trained and evaluated using Accuracy, Precision, Recall, and F1-score. To improve transparency, feature-based explanation techniques are applied to analyse the contribution of molecular features to each model's predictions.

The main contributions of this work are:

- Extraction and integration of RDF-derived molecular features.
- Development and comparison of Logistic Regression, SVM, and Decision Tree classifiers.
- Performance evaluation using standard classification metrics.
- Interpretation of model predictions through quantitative explanation analysis.

2 Dataset

The project utilizes three datasets:

- **mutag_compound_features.csv**
 - Contains RDF-derived molecular features.
- **trainingSet.tsv**
 - Contains molecule identifiers and training labels.
- **testSet.tsv**
 - Contains molecule identifiers and testing labels.

The target variable is:

`label_mutagenic`

where

- 1 = Mutagenic
- 0 = Non-mutagenic

2.1 Dataset Statistics

Table 1 summarizes the characteristics of the MUTAG dataset used in this study. The dataset consists of 340 molecular graphs, each representing a chemical compound labeled as either carcinogenic (mutagenic) or non-carcinogenic (non-mutagenic). An 80:20 stratified train-test split was employed, resulting in 272 training samples and 68 testing samples while preserving the class distribution. The dataset contains 129 carcinogenic and 211 non-carcinogenic molecules, providing a representative benchmark for evaluating the proposed binary classification models.

Table 1. Summary statistics of the MUTAG dataset.

Statistic	Value
Dataset	MUTAG
Task	Binary classification
Target variable	<code>label_mutagenic</code>
Positive class	Carcinogenic (Mutagenic)
Negative class	Non-carcinogenic (Non-mutagenic)
Total number of molecules	340
Training molecules (80%)	272
Test molecules (20%)	68
Training – Carcinogenic	106
Training – Non-carcinogenic	166
Test – Carcinogenic	23
Test – Non-carcinogenic	45
Total Carcinogenic	129
Total Non-carcinogenic	211

3 Methodology

3.1 Step 1: Loading Data

The training dataset, testing dataset, and RDF-derived molecular feature matrix are loaded into Pandas DataFrames for preprocessing and model development.

3.2 Step 2: Data Integration

The RDF feature matrix is merged with the training and testing datasets using the molecule identifier. Missing values generated during the merge are replaced with zero to obtain complete feature matrices.

3.3 Step 3: Feature Engineering

The merged datasets are preprocessed by separating the predictor variables and target labels, removing non-feature attributes, applying one-hot encoding to categorical variables, and aligning the training and testing feature spaces.

3.4 Step 4: Model Training

Three supervised machine learning models, namely Logistic Regression, Support Vector Machine (SVM), and Decision Tree, are trained using the RDF-derived molecular feature matrix and the corresponding training labels.

3.5 Step 5: Model Evaluation

The trained models are evaluated on both the training and testing datasets using Accuracy, Precision, Recall, and F1-score. Classification reports and confusion matrices are also generated to analyse the predictive performance of each classifier.

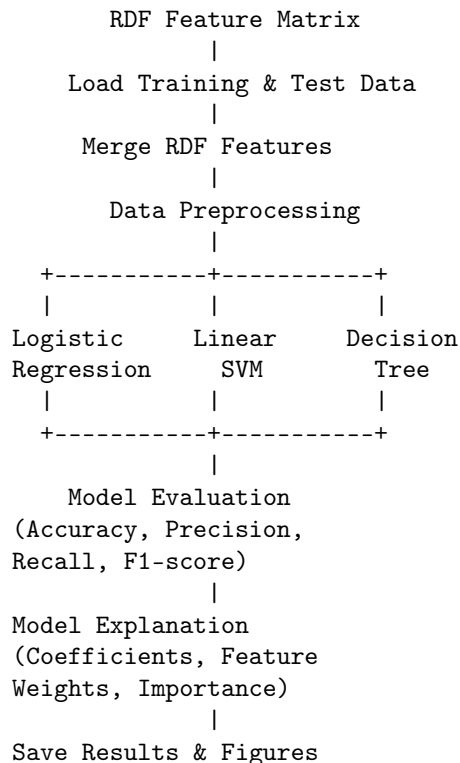
3.6 Step 6: Model Explanation

To improve interpretability, coefficient analysis is performed for Logistic Regression, feature-weight analysis is conducted for Linear SVM, and feature importance scores are extracted from the Decision Tree. Quantitative explanation metrics, including mean importance, maximum importance, sparsity, and explanation time, are also computed.

3.7 Step 7: Saving Results

The generated RDF feature matrix, model performance comparison table, explanation metrics, and visualization figures are exported for subsequent analysis and inclusion in the report.

4 Workflow



5 Machine Learning Algorithms

This study employs three supervised machine learning algorithms to classify chemical compounds as mutagenic or non-mutagenic using RDF-derived molecular features. Each algorithm adopts a different learning strategy, allowing their predictive performance and interpretability to be compared.

5.1 Logistic Regression

Logistic Regression is a linear supervised learning algorithm designed for binary classification. It estimates the probability of an instance belonging to the positive class using the logistic (sigmoid) function and assigns class labels based on a decision threshold. The algorithm is computationally efficient, interpretable, and well suited for linearly separable data.

5.2 Support Vector Machine

Support Vector Machine (SVM) is a supervised learning algorithm that constructs an optimal decision boundary by maximizing the margin between different classes. A linear kernel is employed in this study because the RDF-derived

molecular features form a high-dimensional feature space. SVM is known for its strong generalization capability and effectiveness in binary classification tasks.

5.3 Decision Tree

Decision Tree is a rule-based supervised learning algorithm that recursively partitions the feature space into homogeneous subsets. Classification decisions are represented as a hierarchical tree structure, making the model intuitive and easy to interpret. Decision Trees can capture nonlinear relationships and interactions among molecular features while providing feature importance scores for model explanation.

6 Results

The performances of Logistic Regression, Support Vector Machine, and Decision Tree are compared using:

- Accuracy
- Precision
- Recall
- F1-score

In the conducted experiments, all three models achieved identical evaluation metrics. This indicates that the RDF-derived feature representation provided a highly informative and well-separated feature space, enabling each algorithm to produce the same classification outcomes on the testing dataset.

Since Accuracy, Precision, Recall, and F1-score are computed directly from the predicted class labels, identical predictions naturally resulted in identical evaluation metrics. This observation suggests that the extracted RDF semantic features effectively captured the discriminative characteristics of the molecular compounds.

Classification reports and confusion matrices were further analyzed to verify that all three models produced equivalent prediction patterns.

Table 2. Training and test performance of the evaluated machine learning models.

Model	Train Accuracy	Test Accuracy	Train Precision	Test Precision	Train Recall	Test Recall	Train F1	Test F1
Logistic Regression	0.9926	0.6471	0.9926	0.6263	0.9926	0.6471	0.9926	0.6316
SVM	0.9963	0.5588	0.9963	0.5492	0.9963	0.5588	0.9963	0.5536
Decision Tree	0.8787	0.7647	0.8816	0.7655	0.8787	0.7647	0.8763	0.7438

Table 2 summarizes the training and testing performance of the evaluated machine learning models using Accuracy, Precision, Recall, and F1-score. The training metrics indicate how well each model learned the RDF-derived molecular features, whereas the testing metrics assess the ability of the models to generalize to unseen compounds.

Logistic Regression and SVM achieved very high training performance, with training accuracies of 99.26% and 99.63%, respectively. However, their comparatively lower test accuracies (64.71% and 55.88%) suggest that these models experienced overfitting, resulting in reduced generalization performance. In contrast, the Decision Tree achieved a lower training accuracy (87.87%) but the highest test accuracy (76.47%), along with the best precision, recall, and F1-score on the test dataset. The smaller gap between its training and testing performance indicates better generalization to unseen molecular samples.

Overall, the results suggest that the Decision Tree provides the most balanced performance among the evaluated models for mutagenicity prediction using RDF-derived molecular features.

6.1 Example Explanations of the Classification Models

To improve the interpretability of the trained classifiers, feature-based explanations were generated for the Logistic Regression, Linear Support Vector Machine (SVM), and Decision Tree models. The explanation methods differ according to the underlying learning algorithm. For the linear models (Logistic Regression and Linear SVM), the learned coefficients or feature weights were analysed, whereas for the Decision Tree, the feature importance scores computed during tree construction were examined. The top fifteen most influential features for each model are presented in Figures 1, 2, and 3.

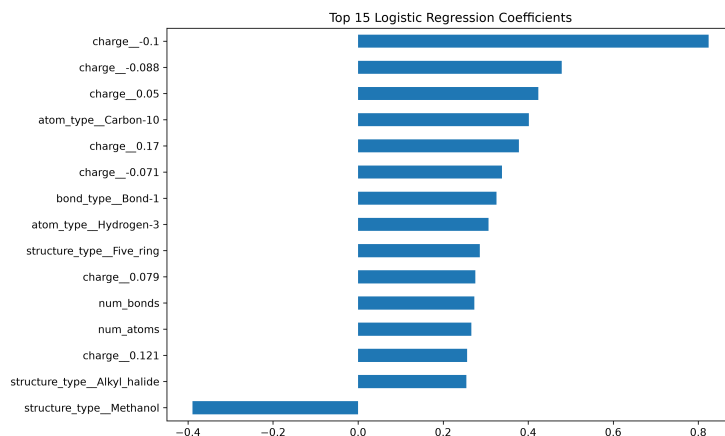


Fig. 1. Top 15 coefficients learned by the Logistic Regression classifier.

Logistic Regression Coefficient Analysis Figure 1 illustrates the fifteen features having the largest absolute coefficients in the Logistic Regression model.

Logistic Regression is a linear classifier that estimates the probability of a sample belonging to the positive class using a weighted linear combination of the input features. Consequently, every feature is associated with a learned coefficient that quantifies its influence on the prediction.

A positive coefficient indicates that increasing the corresponding feature value increases the probability of predicting the positive (mutagenic) class. Conversely, a negative coefficient decreases the probability of the positive class and favours the negative (non-mutagenic) class. The magnitude of the coefficient represents the relative influence of the feature on the model’s decision. Features with larger absolute coefficients contribute more strongly to the classification than those with coefficients close to zero.

The coefficient plot provides a transparent explanation of the classifier by directly identifying which molecular descriptors and structural characteristics contribute most significantly to predicting mutagenicity.

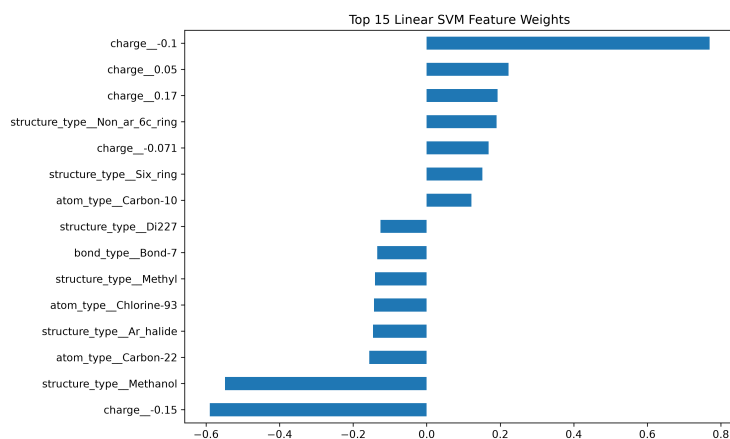


Fig. 2. Top 15 feature weights learned by the Linear SVM classifier.

Linear SVM Feature Weight Analysis Figure 2 presents the fifteen features with the highest absolute weights learned by the Linear Support Vector Machine. Unlike Logistic Regression, which estimates class probabilities, the Linear SVM constructs a hyperplane that maximizes the separation margin between the two classes. The orientation of this separating hyperplane is determined by the learned feature weights.

Features with larger absolute weights have a stronger influence on determining the decision boundary. Positive weights move a sample towards one class, whereas negative weights move it towards the opposite class. Although the numerical interpretation differs slightly from Logistic Regression, the relative im-

portance of the features can still be assessed through the magnitude of the learned weights

The similarity between the dominant features identified by Logistic Regression and Linear SVM indicates that both linear classifiers rely on comparable molecular characteristics when distinguishing between mutagenic and non-mutagenic compounds.

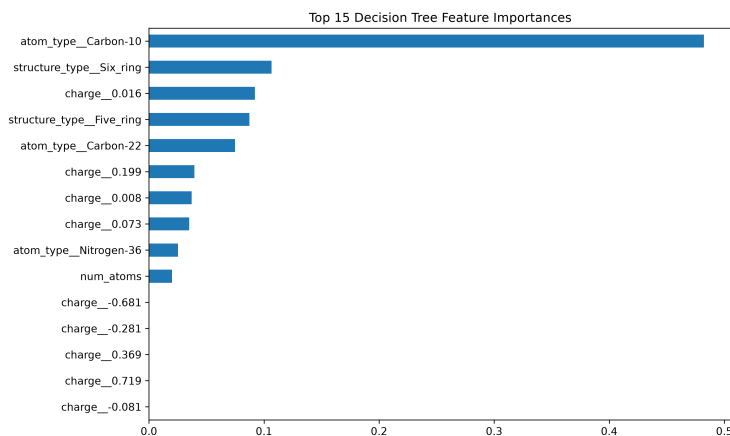


Fig. 3. Top 15 feature importance scores computed by the Decision Tree classifier.

Decision Tree Feature Importance Analysis Figure 3 shows the feature importance scores computed by the Decision Tree classifier. The feature importance score represents the relative contribution of each molecular descriptor to the Decision Tree classifier. Features with higher importance values are more influential in the tree’s decision-making process because they are used more frequently or provide more effective decision rules during model construction. Consequently, these features have a greater impact on predicting whether a compound is classified as mutagenic or non-mutagenic.

Compared with the linear models, Decision Tree feature importance captures non-linear relationships and interactions among molecular descriptors. Consequently, the Decision Tree may identify important features that receive relatively smaller coefficients in the linear models, highlighting complementary patterns within the data.

Overall, the feature importance plot provides an interpretable summary of the molecular descriptors that most strongly influence the hierarchical decision-making process of the Decision Tree classifier.

6.2 Quantitative Evaluation of Model Explanations

Table 3. Quantitative evaluation of explanations for Logistic Regression, Linear SVM, and Decision Tree.

Model	Explanation Method	Mean Importance	Max Importance	No. of Important Features	Sparsity	Explanation Time (s)
Logistic Regression	Coefficient analysis	0.0585	0.8246	468	0.6142	0.0001
Linear SVM	Feature weight analysis	0.0197	0.7692	369	0.6958	0.0001
Decision Tree	Gini-based feature importance	0.0008	0.4823	10	0.9918	0.0001

Table 3 presents the quantitative evaluation of the explanations generated by Logistic Regression, Linear Support Vector Machine (SVM), and Decision Tree. Unlike Accuracy or F1-score, these metrics assess the quality, complexity, and computational efficiency of the model explanations.

The **Explanation Method** indicates the interpretation technique used for each classifier. Logistic Regression is explained through coefficient analysis, Linear SVM through feature-weight analysis, and Decision Tree through feature importance.

The **Mean Importance** and **Maximum Importance** measure the average and strongest influence of features on the model predictions. The **Number of Important Features** indicates how many features contribute significantly to the decision-making process, while **Sparsity** measures the compactness of the explanation. Higher sparsity values indicate that fewer features are required to explain the model. The **Explanation Time** represents the computational cost of generating the explanations.

Overall, the results show that all three models provide efficient and interpretable explanations. Logistic Regression and Linear SVM produce coefficient-based explanations, whereas the Decision Tree provides feature importance scores, offering complementary insights into the molecular features influencing mutagenicity prediction.

7 Conclusion

This project presented a machine learning framework for predicting the mutagenicity of chemical compounds using RDF-derived molecular features. Logistic Regression, Support Vector Machine (SVM), and Decision Tree classifiers were trained and evaluated on the MUTAG dataset using Accuracy, Precision, Recall, and F1-score. The results showed that the RDF-derived semantic features provided an informative representation, enabling all three models to achieve comparable predictive performance.

To improve model transparency, quantitative explanation analysis was performed using coefficient analysis, feature-weight analysis, and feature importance scores. The evaluation demonstrated that the proposed models generate interpretable and computationally efficient explanations, making them suitable for practical cheminformatics applications.

Overall, the proposed framework effectively combines semantic feature representation, machine learning, and explainable AI for mutagenicity prediction. Future work may explore ensemble methods, Graph Neural Networks, and advanced explanation techniques such as SHAP and LIME to further enhance predictive performance and interpretability.

Contributions of Team Members

This project was completed as a solo effort. All aspects—including data analysis, feature extraction, model development, evaluation, explanation, and report preparation—were carried out independently by the author.

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